



PYTHON PROJECT • EXPLORATORY DATA ANALYSIS

Marketing Campaigns Analysis

Examining the marketing mix — product, price, place and promotion — through EDA and hypothesis testing to understand what drives customer acquisition.

Shivani Sharma

Data Analyst | Generative AI



PROBLEM STATEMENT

Decoding the Marketing Mix

Course-end project brief: understand how product, price, place and promotion drive customer acquisition.

SITUATION

As a data analyst, examine variables such as birth year, education, income, product expenditure, sales channels and promotional outcomes to uncover the patterns that guide marketing strategy.

TASK

Run exploratory data analysis and hypothesis testing on the marketing dataset to measure how each of the four Ps influences whether a customer is acquired and retained.

RESULT

A data-driven report that optimizes marketing strategy, refines product offerings and improves customer engagement.



Product



Price



Place



Promotion

Dataset at a Glance

marketing_data.csv loaded with pandas, then inspected for shape, types and completeness.



2,240

Customer records



28

Original columns



24

Missing incomes
found



5

Columns engineered
for analysis



Key fields used in this analysis

Year_Birth, Education, Marital_Status, Income, Kidhome/Teenhome, Dt_Customer, Recency, six spending categories (MntWines ... MntGoldProds), four purchase-channel counts, campaign-acceptance flags (AcceptedCmp1–5, Response) and Country.

Fixing Income & Dates

Two columns needed type correction before any analysis could run.



Income → float

```
data["Income"].str.replace("$", "").str.replace(",", "").  
astype(float)
```

Stored as text like “\$84,835.00”, Income is stripped of currency symbols and commas, then cast to a numeric type so it can be aggregated and plotted.



Dt_Customer → datetime

```
data["Dt_Customer"] =  
pd.to_datetime(data["Dt_Customer"])
```

Enrollment dates arrive as plain text; converting to a proper datetime type unlocks date arithmetic and time-based grouping later in the analysis.

DATA QUALITY

Cleaning Categories & Imputing Income

24 customers were missing income — filled using peers in the same education and marital group, not a blunt overall average.

CATEGORY CLEAN-UP

Education: “2n Cycle” → Master, “Basic” → Undergraduate

Marital status: Together → Married; Divorced, Widow, Alone, YOLO, Absurd → Single

Result: 8 marital labels and 5 education labels collapsed into 2 and 4 clean, analysis-ready categories.

Married

1,444

Single

796



Group-Mean Imputation

```
group_mean =  
data.groupby(["Education", "Marital_Status"])["Income"].trans  
form("mean")
```

Each missing Income is filled with the average income of customers sharing the same education level and marital status — a far more realistic estimate than a single global average.

Result: 0 missing values remain in Income.

Five New Variables for Deeper Analysis

Raw columns transformed into the metrics the rest of the analysis runs on.



Age

2014 – Year_Birth



Spending

Sum of 6 product categories
(wines, fruits, meat, fish, sweets, gold)



Children_count

Kidhome + Teenhome



has_children

1 if Children_count > 0, else 0



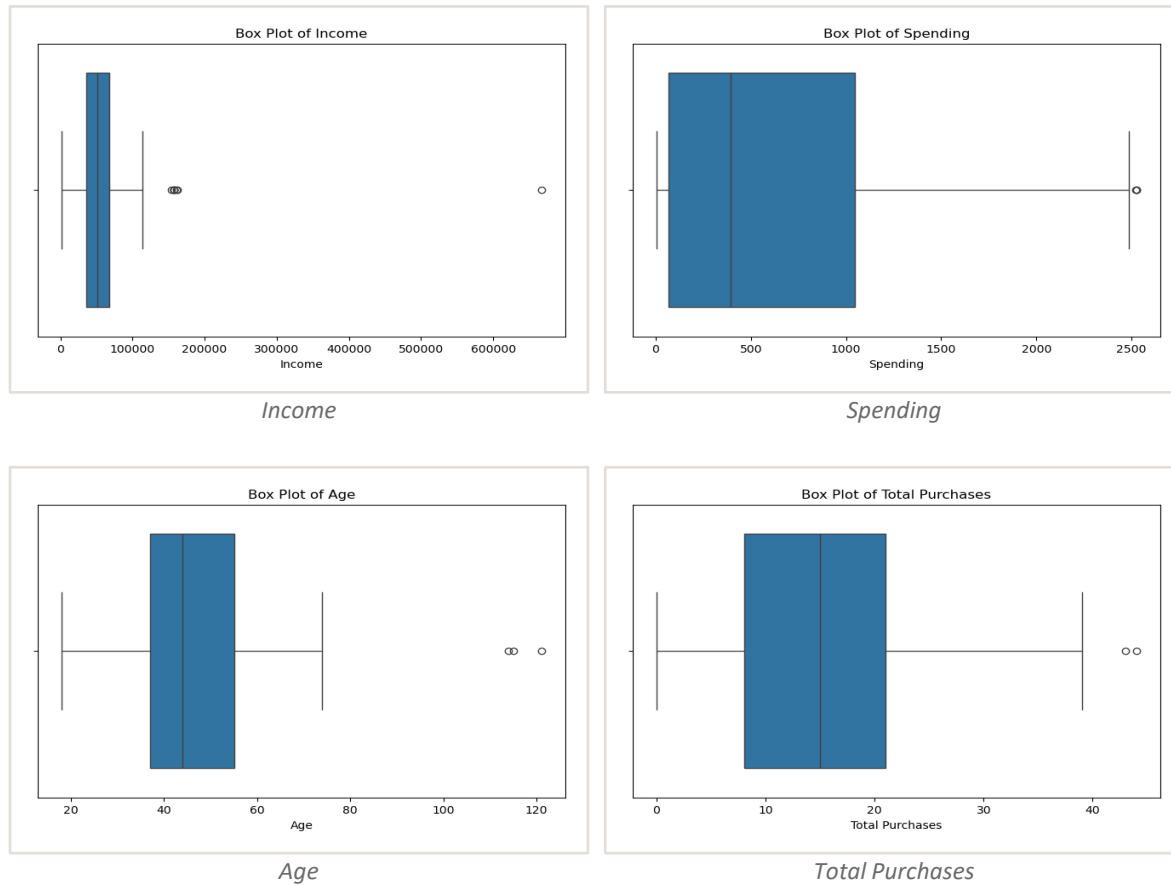
total_purchases

Deals + Web + Catalog + Store
purchase counts

OUTLIER DETECTION

Box Plots Reveal Extreme Values

IQR-based bounds computed for Income, Spending, Age and Total Purchases.



IQR BOUNDS (1.5× RULE)

Income

Lower -13,588 → Upper 117,416

Spending

Lower -1,396 → Upper 2,511

Age

Lower 10 → Upper 82

Total purchases

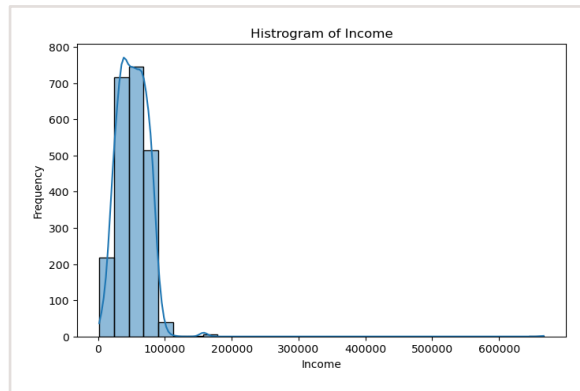
Lower -11.5 → Upper 40.5

All four columns capped to these limits — extreme values pulled in, no rows dropped.

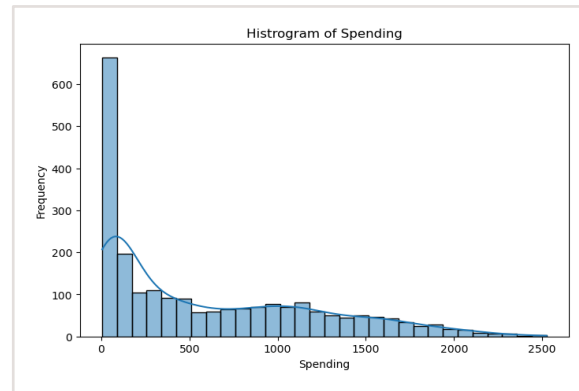
DISTRIBUTION ANALYSIS

How the Core Metrics Are Distributed

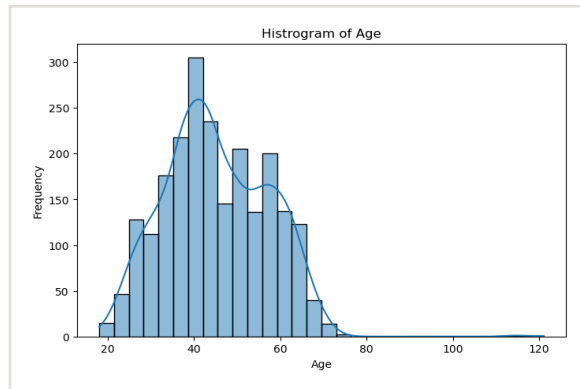
Income averages \$52,253, but a max of \$666,666 before capping shows why outlier treatment mattered.



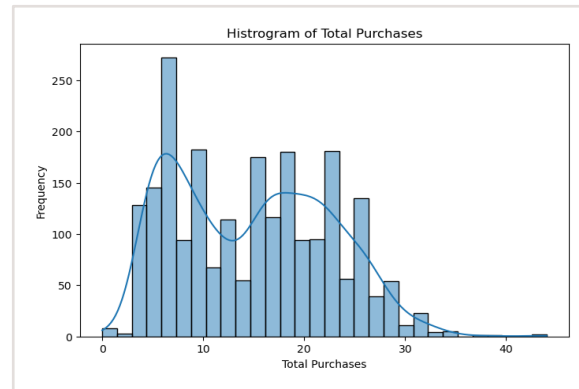
Income



Spending



Age



Total Purchases

INCOME, POST-CLEANING

Mean	\$52,253
Median	\$51,533
Std Dev	\$25,039
Min	\$1,730
Max (raw)	\$666,666

Income and Spending are right-skewed with a long tail of high-value customers; Age is close to a normal spread across the customer base.

Turning Categories into Numbers

Two encoding strategies chosen to match the nature of each categorical field.



Ordinal Encoding — Education

Education carries a natural order, so each level maps to a rank:

Undergraduate	1
Graduation	2
Master	3
PhD	4



One-Hot Encoding

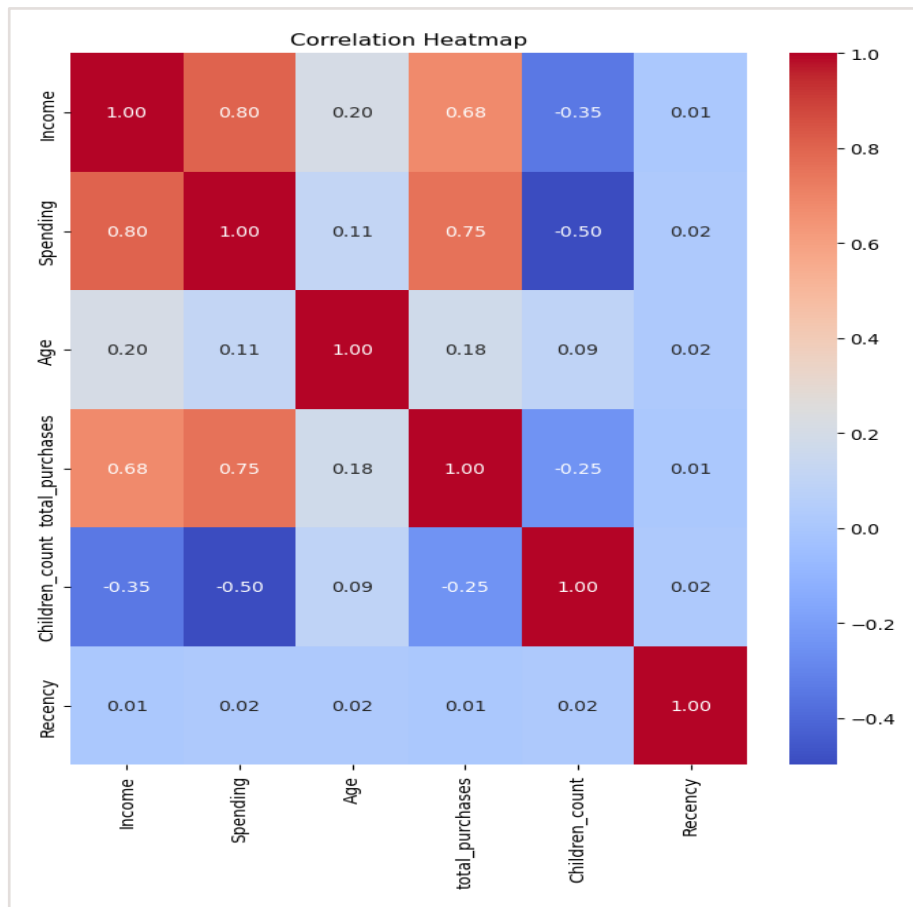
Country and Marital_Status have no inherent order, so each category becomes its own binary column via `pd.get_dummies()`, dropping the first level to avoid redundancy.

```
data = pd.get_dummies(data, columns=["Marital_Status"],  
drop_first=True)  
data = pd.get_dummies(data, columns=["Country"],  
drop_first=True)
```

CORRELATION ANALYSIS

What Moves Together?

A six-variable correlation matrix highlights the strongest relationships in customer behavior.



Income ↔ Spending

+0.80

Higher-income customers spend substantially more.

Spending ↔ Total Purchases

+0.75

Bigger spenders also transact more often.

Children_count ↔ Spending

-0.50

Households with more children spend noticeably less.

Recency ↔ everything else

≈ 0.00

Days since last purchase shows almost no linear relationship with any other metric.

HYPOTHESIS TESTING

Does Spending Predict Campaign Acceptance?

Welch's t-test comparing Spending between customers who accepted Campaign 5 and those who did not.

H_0 (null)

Mean spending is the same for customers who accepted Campaign 5 and those who didn't.

H_1 (alternative)

Mean spending differs between the two groups — tested with `stats.ttest_ind(equal_var=False)`.

T-STATISTIC

31.30

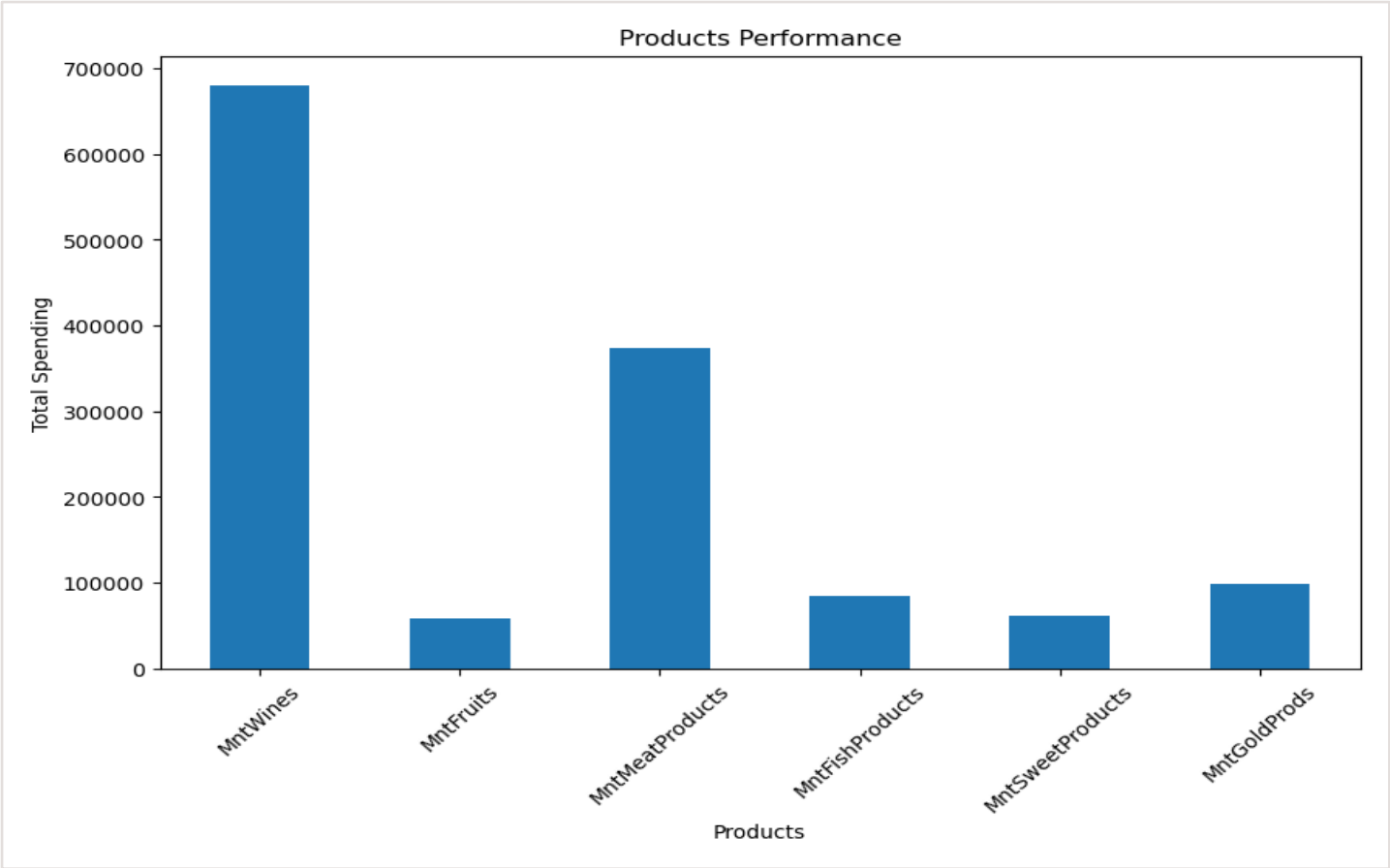
P-VALUE

2.12×10^{-80}

Conclusion: reject H_0 . The p-value is far below 0.05 — spending is a strong, statistically significant predictor of whether a customer accepts a targeted campaign.

Wine Leads Every Product Category

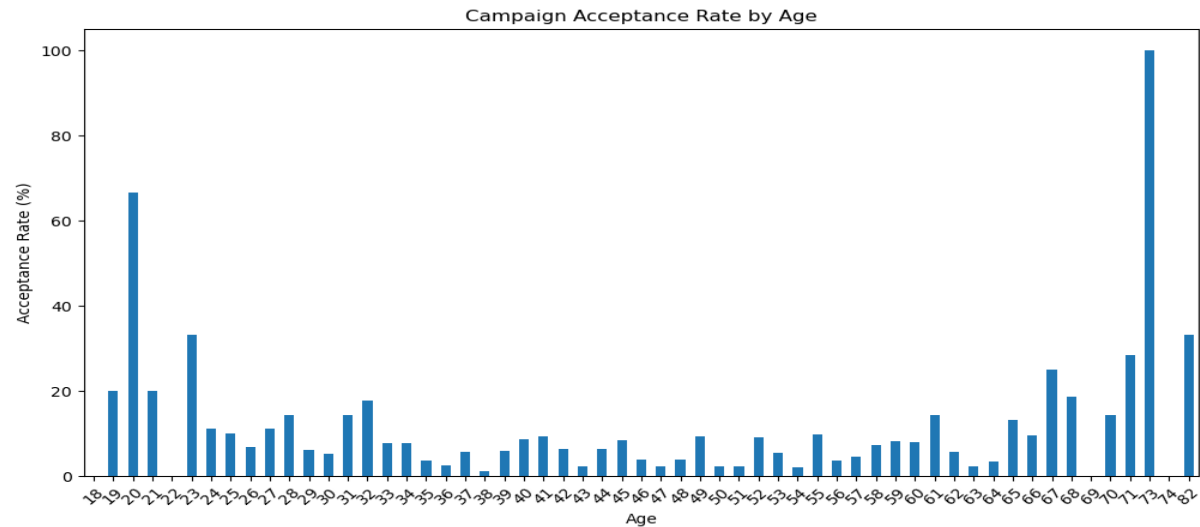
Total spending across the six product lines, summed over all 2,240 customers.



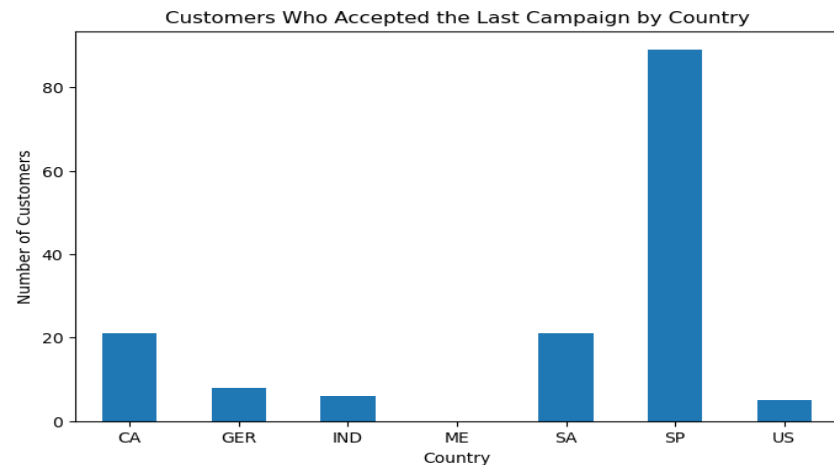
Wines	680,816
Meat Products	373,968
Gold Products	98,609
Fish Products	84,057
Sweet Products	60,621
Fruits	58,917

Who Responds to Campaigns?

Campaign 5 acceptance rate by customer age, and last-campaign acceptance by country.



Acceptance rate by age — sparse older cohorts show volatile spikes



Spain dominates

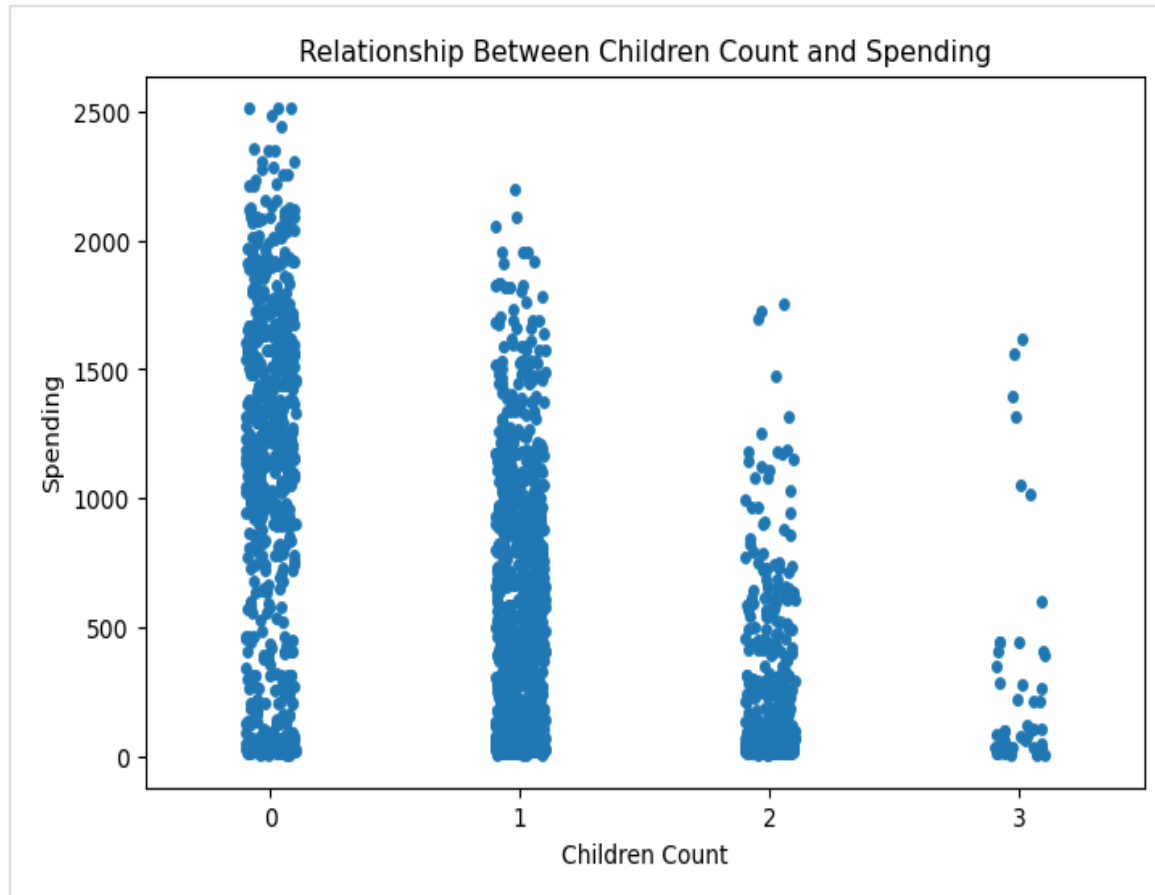
89

customers accepted the last campaign — more than 4× the next-highest countries (Canada and South Africa, 21 each).

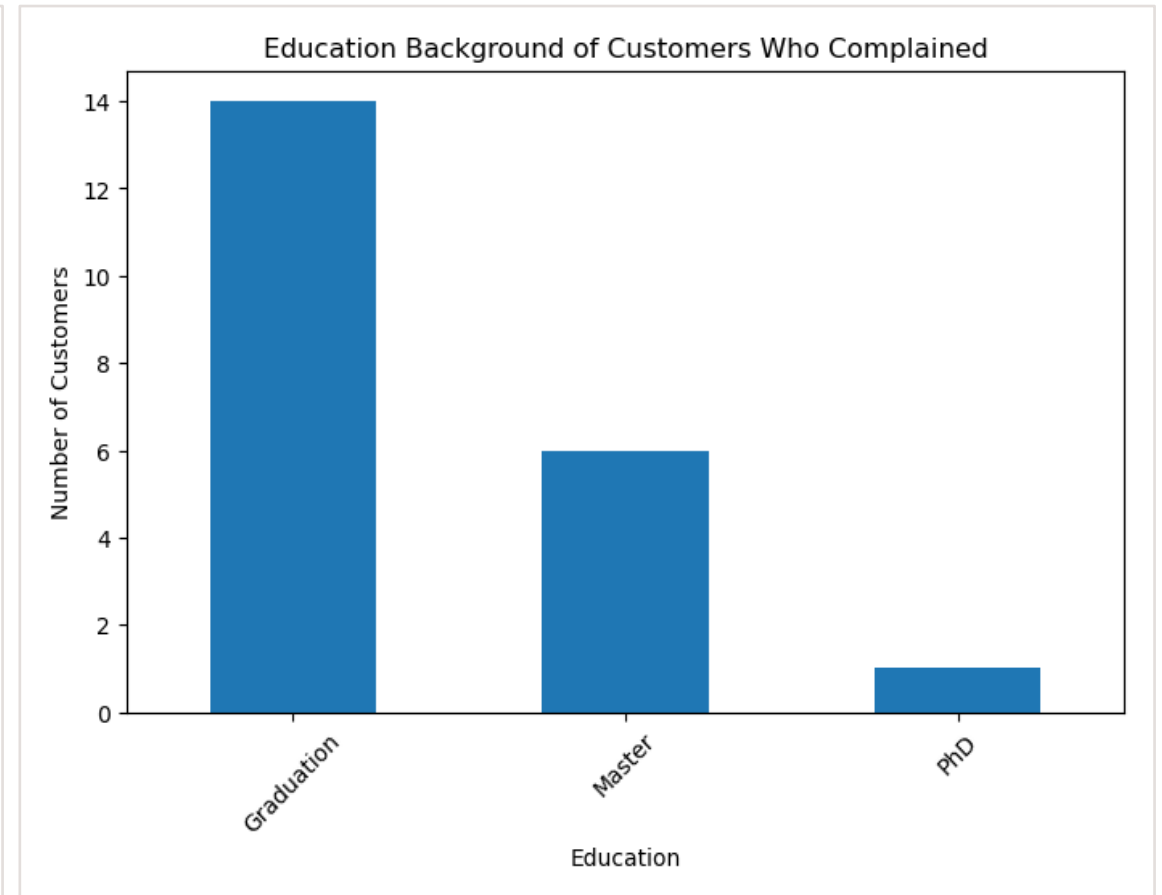
Germany, India and the US trail well behind; Mexico recorded zero acceptances in this campaign.

Family Size, Complaints & Spend

Children in the household correlate with lower spend; complaints skew toward Graduation-level customers.



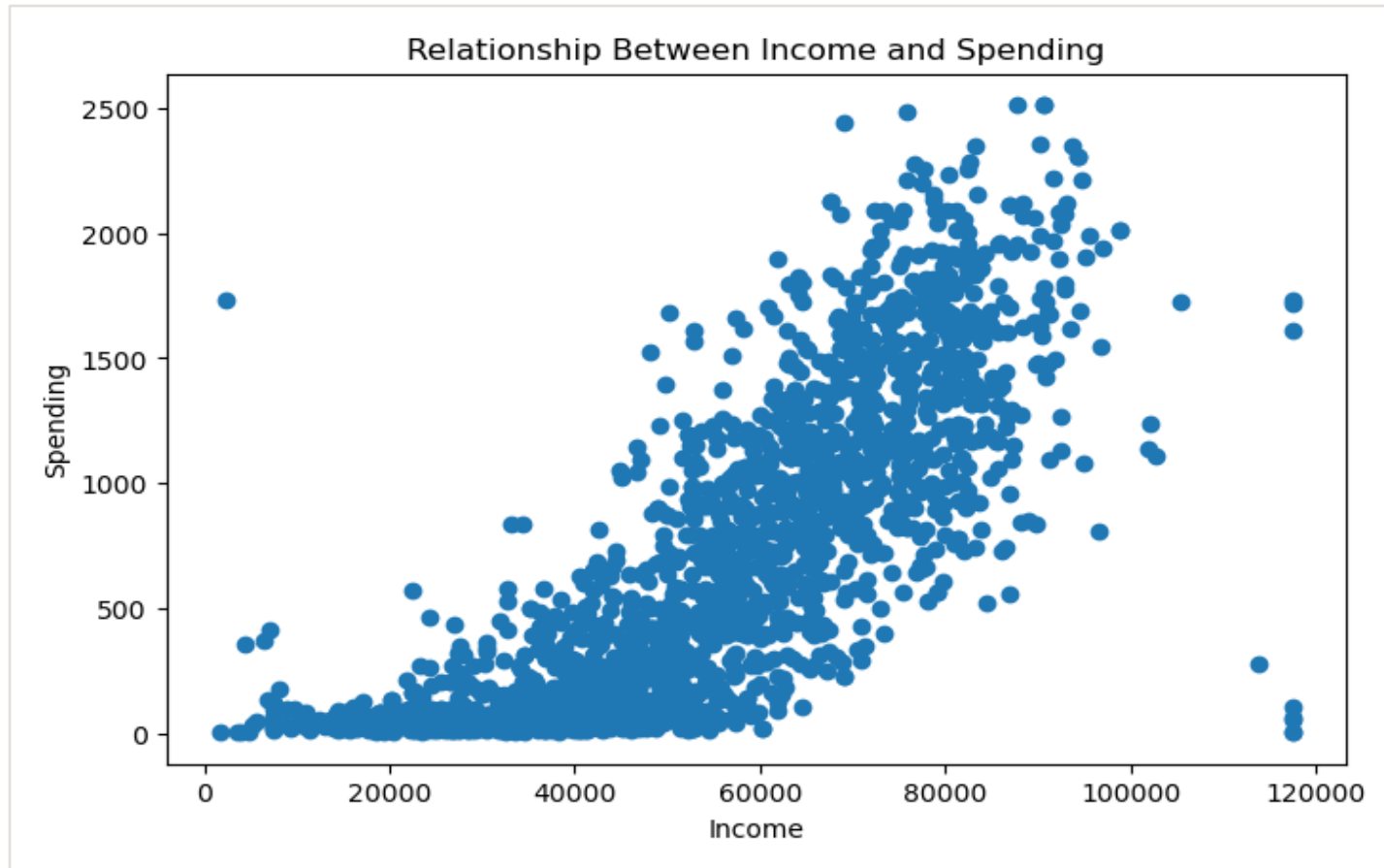
Spending drops sharply as children count rises



14 of 21 complaints came from Graduation-level customers

Income Is the Strongest Lever on Spending

A clear upward relationship, consistent with the +0.80 correlation seen earlier.



READING THE SCATTER

Below roughly \$40,000 in income, spending stays flat and low regardless of how much a customer earns. Past that point, spending rises steeply and consistently with income — a natural break-point for tiered marketing offers.

A handful of high-income, low-spend outliers remain — candidates for re-engagement campaigns.

KEY INSIGHTS

What the Data Tells Marketing



Spending predicts campaign response

$T = 31.30$, $p \approx 2.1 \times 10^{-80}$ — target high-spend segments for future promotions.



Income is the strongest driver of spend

+0.80 correlation, with a break-point near \$40K — tier offers around this threshold.



Wine and meat lead product spend

Together over 74% of total product revenue — prioritize inventory and promotion here.



Children reduce discretionary spend

-0.50 correlation with Spending — family-sized bundles may resonate more than premium upsells.



Spain is the core acceptance market

89 accepted the last campaign, 4× the next country — concentrate promotional budget geographically.

Thank You

Questions on the Marketing Campaigns Analysis project are welcome.

Shivani Sharma

Data Analyst | Generative AI

shivani92754@gmail.com

+91 82945 99916

[linkedin.com/in/shivani-sharma-517636413](https://www.linkedin.com/in/shivani-sharma-517636413)